

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                |    |    | Marks available |     |     |       |       |      |    |  |  |   |  |   |  |  |
|----------|-----|-------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----------------|-----|-----|-------|-------|------|----|--|--|---|--|---|--|--|
|          |     |       |  |                                                                                                                                                                                                                                                                                                |    |    | AO1             | AO2 | AO3 | Total | Maths | Prac |    |  |  |   |  |   |  |  |
| 4        | (a) | (i)   |  | <table><tr><td></td><td>B</td><td>b</td></tr><tr><td>B</td><td>BB</td><td>Bb</td></tr><tr><td>b</td><td>Bb</td><td>bb</td></tr></table>                                                                                                                                                        |    | B  | b               | B   | BB  | Bb    | b     | Bb   | bb |  |  | 2 |  | 2 |  |  |
|          |     |       |  |                                                                                                                                                                                                                                                                                                | B  | b  |                 |     |     |       |       |      |    |  |  |   |  |   |  |  |
|          |     |       |  | B                                                                                                                                                                                                                                                                                              | BB | Bb |                 |     |     |       |       |      |    |  |  |   |  |   |  |  |
|          |     |       |  | b                                                                                                                                                                                                                                                                                              | Bb | bb |                 |     |     |       |       |      |    |  |  |   |  |   |  |  |
|          |     |       |  |                                                                                                                                                                                                                                                                                                |    |    |                 |     |     |       |       |      |    |  |  |   |  |   |  |  |
|          |     |       |  |                                                                                                                                                                                                                                                                                                |    |    |                 |     |     |       |       |      |    |  |  |   |  |   |  |  |
|          |     |       |  |                                                                                                                                                                                                                                                                                                |    |    |                 |     |     |       |       |      |    |  |  |   |  |   |  |  |
|          |     | (ii)  |  | Circle around <b>bb</b>                                                                                                                                                                                                                                                                        |    |    | 1               |     | 1   |       |       |      |    |  |  |   |  |   |  |  |
|          |     | (iii) |  | The Punnett square shows the { <u>probability/ chance/ possibility</u> } of {the different genotypes / each outcome}/ shows the possible genotypes (1)<br>not {the number of/ the amount of/ how many} offspring (1)                                                                           |    |    | 2               |     | 2   | 1     |       |      |    |  |  |   |  |   |  |  |
|          | (b) |       |  | Any <b>one</b> (×1) from:<br>(Scientific names/ they) are {universal/ same all over the World (1)<br>Common names are {different all over the World/different in different languages} / avoids duplication with {local/ common} names (1)<br>Avoids confusion with the use of common names (1) | 1  |    |                 |     | 1   |       |       |      |    |  |  |   |  |   |  |  |

| Question         |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks available |     |     |       |       |      |
|------------------|-----|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|                  |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | AO1             | AO2 | AO3 | Total | Maths | Prac |
|                  | (c) |  |  | <ul style="list-style-type: none"><li>Polar bears live {in the Arctic (Circle)/in very cold climates/in colder climates} /ORA/ Polar bears live at -8 °C and black bears live 12°C (1)</li></ul> <p>Any <b>one</b> (×1) from:</p> <ul style="list-style-type: none"><li>The {larger the mass/ smaller sa:vol ratio} {the less heat is lost / more heat is conserved/ the less heat the body loses} (1)</li><li>Polar bear may have {a higher proportion of/ thicker layer of/ more} {fat/blubber} (1)<br/>ORA for NA black bear (ignore any ref. to hibernation/ fur)</li></ul> |                 | 1   | 1   | 2     |       |      |
| Question 4 total |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1               | 6   | 1   | 8     | 1     | 0    |